

Neural Controlled Differential Equations for Irregular Time Series

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Motivation

How to choose model for [sequential data](#)?

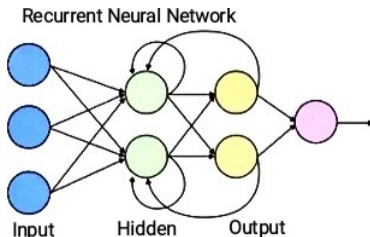
- **Recurrent Neural Networks**

Interpretation: discrete approximation to some function of process.

Problem: discretisation breaks down if the data is irregularly sampled or partially observed.

- **Neural Differential Equations**

Motivation: consider models in continuous time.



Neural Ordinary Differential Equations

Neural ODE model

Approximate a map $x \rightarrow y$ by learning f_θ and linear maps l_θ^1, l_θ^2 :

$$y \approx l_\theta^1(z_T), \quad z_t = z_0 + \int_0^t f_\theta(z_s) ds, \quad z_0 = l_\theta^2(x).$$

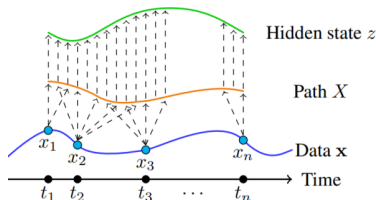
Problems:

- Not involve sequential data, time dimension t integrated over;
- For learned θ the solution is determined by the initial condition. There is no direct mechanism for incorporating late-arriving data.

Neural Controlled Differential Equation

- Time series $x = ((t_0, x_0), (t_1, x_1), \dots, (t_n, x_n))$, $t_i \in \mathbb{R}$, $x_i \in \mathbb{R}^\nu$;
- Cubic spline $X : [t_0, t_n] \rightarrow \mathbb{R}^{\nu+1}$, with $X_{t_i} = (x_i, t_i)$;
- Neural networks $f_\theta : \mathbb{R}^\omega \rightarrow \mathbb{R}^{\omega \times (\nu+1)}$ and $\zeta_\theta : \mathbb{R}^{\nu+1} \rightarrow \mathbb{R}^\omega$;
- Neural CDE model

$$z_t = z_{t_0} + \int_{t_0}^t f_\theta(z_s) dX_s, \quad z_{t_0} = \zeta_\theta(x_0, t_0), \quad t \in (t_0, t_n].$$



Patrick Kidger, et al. Neural Controlled Differential Equations for Irregular Time Series // NIPS 2020.

Evaluating the Neural CDE model

Notation: $g_{\theta, X}(x, s) := f_{\theta}(z) \frac{dX}{ds}(s)$.

For $t \in (t_0, t_n]$

$$\begin{aligned} z_t &= z_{t_0} + \int_{t_0}^t f_{\theta}(z_s) dX_s = z_{t_0} + \int_{t_0}^t f_{\theta}(z_s) \frac{dX}{ds}(s) ds = \\ &= z_{t_0} + \int_{t_0}^t g_{\theta, X}(x, s) ds. \end{aligned}$$

Possible to solve the Neural CDE the same way as Neural ODEs.

Experiment 1: irregularly sampled time series

CharacterTrajectories dataset: pen position and force during single-stroke Latin character writing.

Goal: to classify which of 20 different characters are written.

Table 1: Test accuracy (mean $\pm$ std, computed across five runs)

Model	Test Accuracy		
	30% dropped	50% dropped	70% dropped
GRU-ODE	92.6% $\pm$ 1.6%	86.7% $\pm$ 3.9%	89.9% $\pm$ 3.7%
GRU- Δt	93.6% $\pm$ 2.0%	91.3% $\pm$ 2.1%	90.4% $\pm$ 0.8%
GRU-D	94.2% $\pm$ 2.1%	90.2% $\pm$ 4.8%	91.9% $\pm$ 1.7%
ODE-RNN	95.4% $\pm$ 0.6%	96.0% $\pm$ 0.3%	95.3% $\pm$ 0.6%
Neural CDE (ours)	98.7% $\pm$ 0.8%	98.8% $\pm$ 0.2%	98.6% $\pm$ 0.4%

Experiment 2: regularly sampled time series

SpeechCommands dataset: one-second audio recordings of both background noise and spoken words.

Table 2: Test Accuracy (mean $\pm$ std, computed across five runs)

Model	Test Accuracy
GRU-ODE	47.9% $\pm$ 2.9%
GRU- Δt	43.3% $\pm$ 33.9%
GRU-D	32.4% $\pm$ 34.8%
ODE-RNN	65.9% $\pm$ 35.6%
Neural CDE (ours)	89.8% $\pm$ 2.5%

Adjoint Equations

Neural ODE model:

$$y \approx l^1(z_T, \theta), \quad z_t = z_\tau + \int_\tau^t f(s, z_s, \theta) ds, \quad z_\tau = l^2(x, \theta).$$

Backpropagate integral via [adjoint equations](#) with respect to scalar loss L :

$$a_z(T) = \frac{dL}{dz_T}, \quad a_\theta(T) = 0, \quad a_t(T) = \frac{dL}{dT},$$

$$a_z(t) = a_z(T) - \int_T^t a_z(s) \frac{\partial f}{\partial z}(s, z_s, \theta) ds,$$

$$a_\theta(t) = a_\theta(T) - \int_T^t a_z(s) \frac{\partial f}{\partial \theta}(s, z_s, \theta) ds,$$

$$a_t(t) = a_t(T) - \int_T^t a_z(s) \frac{\partial f}{\partial s}(s, z_s, \theta) ds,$$

$$\frac{dL}{dz_\tau} = a_z(\tau), \quad \frac{dL}{d\theta} = a_\theta(\tau), \quad \frac{dL}{d\tau} = a_t(\tau).$$

Faster ODE Adjoint

Previous methods: take too many steps during backpropagation.

Faster ODE Adjoint^{*}: reduces steps by a median of 40%.

Key observation: a_θ does not appear anywhere in the vector fields. The integral corresponding to a_θ is just an integral, **not an ODE!**

$$a_\theta(T) = 0, \quad a_\theta(t) = a_\theta(T) - \int_T^t a_z(s) \frac{\partial f}{\partial \theta}(s, z_s, \theta) ds, \quad \frac{dL}{d\theta} = a_\theta(\tau).$$

Solving adjoint equations, different accuracy is required for a_z and a_θ .

Semi-norm in error calculation handles non-equivalence.

^{*}Patrick Kidger, et al. "Hey, that's not an ODE": Faster ODE Adjoint via Seminorms // PMLR 2021.

Experiment 3: number of steps in Neural CDE

Table 3: Results for SpeechCommands dataset.

Default norm		Seminorm	
Accuracy (%)	Bwd. NFE (10^6)	Accuracy (%)	Bwd. NFE (10^6)
92.6 $\pm$ 0.4	14.36 $\pm$ 1.09	92.5 $\pm$ 0.5	8.67$\pm$1.60

Figure 1: Mean $\pm$ standard deviation for number of steps.

